

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1581

Levitt 138 kV
April 17, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1581. The Interconnection Customer proposes a 241.2 MW wind facility interconnecting at the Levitt 138 kV substation in the Maryland area. The request was received on October 02, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 476 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$267,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
PRIEST BRIDGE - QUINCEY 138.0 kV Line	CENTRAL AVENUE - QUINCEY 138.0 kV	124.4	422	17.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
PRIEST BRIDGE - QUINCEY 138.0 kV Line	Rebuild line with higher-capacity conductor	\$3,413,000	8% (\$267,000)
	Total allocated to project		\$267,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.